

MGTF 422: Derivatives and Structured Finance

Winter 2026

Monday from 2pm to 4:50pm

Mitch Warachka

mwarachka@ucsd.edu

Office Hours by appointment

COURSE DESCRIPTION

The course provides a comprehensive study of forwards, futures, options, and swaps in equity, currency, and fixed income markets. The pricing of these derivative contracts as well as their applications to risk management are covered in detail. Hedge parameters for options such as Delta and Gamma will be studied along with the binomial option pricing framework. In addition, the course will cover material related to collateralized debt obligations (CDOs) and other structured products. The course will consist primarily of lectures with a generous supply of numerical examples to reinforce important concepts.

LEARNING OBJECTIVES

At the end of this course, you will be able to:

- Understand the concept of No Arbitrage is the foundation for pricing derivatives
- Understand how pricing derivatives arises from replicating their payoffs
- Understand how hedging enables the risk-neutral pricing of derivatives
- Understand how best to choose derivatives for risk management

MATERIALS

No textbook is required. The class covers topics in CFA levels I through III regarding derivatives using a variety of sources.

SCHEDULE

Session	Date	Class Topic & Activities	Assignments
Week 1	January 5 th	Forwards and Futures	
Week 2	January 12 th	Applications to Commodities	
Week 3	January 19 th	No Class	First Assignment: Due January 23 rd
Week 4	January 26 th	Interest Rate and Currency Swaps	
Week 5	February 2 nd	Options	Take-Home Midterm: Due February 6 th
Week 6	February 9 th	Black Scholes and the Greeks	
Week 7	February 16 th	No Class	Second Assignment: Due February 20 th
Week 8	February 23 rd	Binomial Pricing	
Week 9	March 2 nd	Fixed Income Derivatives	
Week 10	March 9 th	Review	Third Assignment: Due March 13 th
Final Exam	March 16 th 3pm -6pm	In person exam	

DELIVERABLES

I – Independent, individual work only. No collaboration or consultation allowed.

C – Collaboration with classmates is allowed. However, each student must submit an individual assignment.

Three Assignments (C)

Two Exams (I)

Class Participation / Attendance (I)

GRADING

Deliverables	Percentage
Participation	10
Assignments (3)	30
Midterm Exam	30
Final Exam	30
Total	100

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:
<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the OSD website for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.